Seminar 1 - Lectures 1, 2 and 3
Computer Vision 1, Master AI, 2026

Slides and exercises inherited from Arnoud Visser, Martin Oswald, Qi Bi, Roan van Blanken

1 Exercise 1: Pinhole Camera

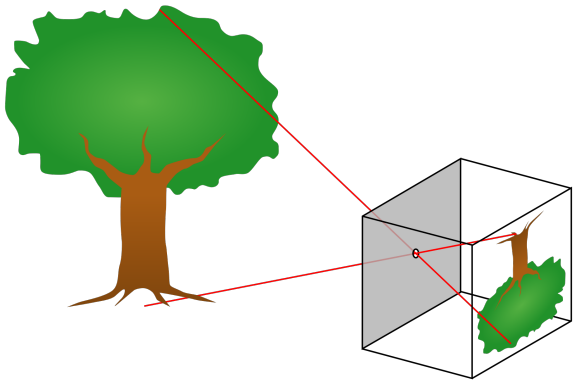


Figure 1: Pinhole camera model (source: Wikipedia)

The camera obscura or pinhole image leverages a natural optical phenomenon. The camera has the shape of a box, light from an object enters through a small hole (the pinhole) in the front and produces an image on the back camera wall.

Q.1.a Annotate the Figure 1 with the focal length, center of projection and image plane. What is the virtual image in this context, and where would you place it?

Q.1.b Following part (a), consider the 3D Euclidean coordinates where the pinhole is located at the origin $\mathbf{O} = [0, 0, 0]^\top$. The Z-axis aligns with the camera axis and points outward from the camera. The Y-axis points vertically upward, and the X-axis points horizontally to the left. Note that this forms a right-handed coordinate system. Draw and label these axes on the provided figure.

Q.1.c Given a focal length $f = 15$, calculate the coordinates $\mathbf{p}$ of the projection on the image plane for the point $\mathbf{P} = [30, 15, 45]^\top$. Additionally, determine the coordinates $\mathbf{p}_v$ on the virtual image plane.

2 Exercise 2: Image Representation



Figure 2: First picture to be scanned, stored and recreated in digital pixels (source: Wikipedia)

A digital image is an image composed of picture elements (pixels). Each pixel has a finite and discrete numeric quantity representing the local intensity. In practice, the set of quantities is determined by the number of bits available per pixel. Depending on whether the image resolution is fixed, it may be a vector or raster type. Here we focus on the latter.

Q.2.a Consider a function $f : X \mapsto Y$. How are digital images represented as a function f? What are common sets for Y and their associated image types?

Q.2.b What are the differences between changes to the quantization level and changes to the spatial resolution? How do these changes affect the visual perception of the image?

3 Exercise 3: Camera Projection

Homogeneous Coordinates

Q.3.a Use the directions defined in Exercise 1, convert the following Cartesian 2D vector into homogeneous coordinates: $\mathbf{v} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

Q.3.b Convert the following homogeneous coordinate into Euclidean coordinates.

$$\mathbf{v}_h = \begin{pmatrix} 6 \\ 9 \\ 12 \\ 3 \end{pmatrix}$$

Q.3.c In the lecture, the extrinsic matrix $[R \,|\, t]$ is discussed. When projecting a 3D point onto a 2D image, what are the dimensions of the matrix $[R \,|\, t]$? Additionally, what are the dimensions of the matrices R and t?

Q.3.d Explain why homogeneous coordinates are necessary.

Camera Translations and Rotations

Assume the camera frame initially coincides with the world frame (i.e., $R = I$, $T = 0$). For each of the following transformations, applied to the camera starting from this initial pose, we aim to write the extrinsic matrix that maps points from world coordinates to camera coordinates.

Q.3.e Write the homogeneous 3×4 matrix for the following transformation: The camera is translated by $+7$ units in the X direction (relative to the world frame).

Q.3.f Write the homogeneous 3×4 matrix for the following transformation: The camera is rotated by 30 degrees around the X-axis (relative to the world frame), with the rotation being clockwise when looking along the positive direction of X.

Q.3.g Write the homogeneous 3×4 matrix for the combination of the rotation followed by the translation as described above.

4 Exercise 4: Reflection

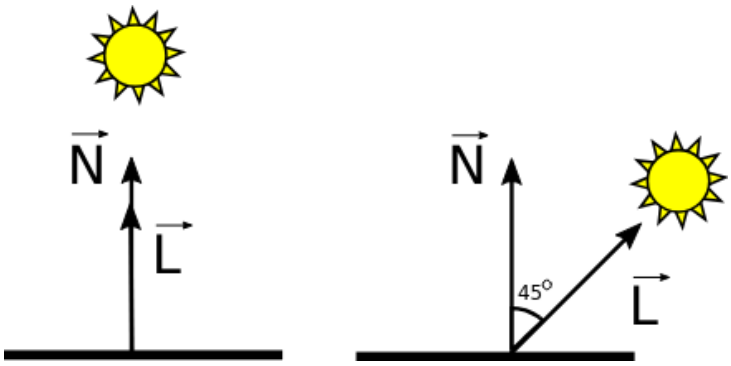


Figure 3: Light source position scenarios: aligned with the surface normal (left) and at an angle of 45° (right).

Consider the color of a matte, non-glossy surface under white light illumination. The color at a specific point on the surface is described by the following reflection model:

$$R = Ik_R \cos\theta, \quad G = Ik_G \cos\theta, \quad B = Ik_B \cos\theta$$

where:

- I is the intensity of the white light source,
- k_R, k_G, and k_B represent the fractions of red, green, and blue light reflected by the surface (i.e., the inherent color of the surface),
- $\cos\theta = \vec{n} \cdot \vec{l}$ is the dot product of two unit vectors: $\vec{n}$ (the surface normal) and $\vec{l}$ (the direction of the light source).

Refer to Figure 3 for visual scenarios where the light source is either aligned with the surface normal or at an angle of 45° to it.

Q.4.a Assume that the surface is flat and homogeneously colored. Explain why the intensity is higher when the surface normal coincides with the direction of the light source compared to when it is observed at an angle relative to the light source direction.

Q.4.b Assume that the color of the surface is yellow, i.e., $R = 100$, $G = 100$, and $B = 10$. Explain what will happen to the values of R, G, and B if the intensity of the light source diminishes. Assume that the camera response to intensity is linear and that intensity is linearly coded.

Q.4.c In the case of a curved (non-flat) surface, assuming no occlusion or interreflection, indicate where the colors will be positioned in the RGB-color space. Explain your answer.

Q.4.d A simple color invariant is given by R/G. Prove that R/G is independent of the intensity of the light source I, object geometry, and the direction of the light source.

Q.4.e Consider the same surface, but assume it is glossy (instead of matte). The reflection model is now given by:

$$R = Ik_R \cos\theta + Ik_s \cos^n\alpha, \quad G = Ik_G \cos\theta + Ik_s \cos^n\alpha, \quad B = Ik_B \cos\theta + Ik_s \cos^n\alpha$$

where k_s is the specular reflection coefficient, and $\cos^n\alpha$ depends on the glossiness and viewing angle.

Prove that $\dfrac{R-G}{R-B}$ is a color invariant for shiny surfaces.

5 Exercise 5: Histograms & Images

Histograms
To calculate the histogram equalization of an image, we use the histogram of the image, binned by gray levels (L). Consider the distribution of pixel values in a 3-bit grayscale image given in Table 1. The total number of gray levels is 8 ($2^3 = 8$), ranging from 0 to 7.

Gray Level	0	1	2	3	4	5	6	7
No. of Pixels	900	800	700	600	500	400	300	200

Table 1: Pixel distribution of a random 3-bit grayscale image

Q.5.a Draw the histogram of the given 3-bit gray levels. What can you infer from the given gray level information? What type of image do you think this represents? Can you reconstruct the image using just the information given in Table 1? Justify your answer.

Q.5.b Calculate the probabilities of pixels for each gray level.

Q.5.c Equalize the histogram and create a table showing the new gray levels and their corresponding pixel counts. Then, draw the equalized histogram.

Images

1	1	1	3	3	4	4	1
4	0	0	5	1	2	0	1
0	1	3	1	2	0	2	0
3	1	2	2	2	1	1	2
1	4	2	2	2	3	5	3
0	2	1	0	2	1	3	0
0	2	1	1	0	0	4	1
0	4	0	0	0	2	1	1

Table 2: 3-bit grayscale image patch

Q.5.d Consider the 3-bit grayscale image patch given in Table 2. Calculate the pixel distribution like in Table 1 and draw the histogram.

Q.5.e Equalize the histogram and then back-project the equalized histogram to obtain the new image. Draw 3-bit grayscale image patch after histogram equalization

Q.5.f So far, you have equalized only grayscale images. How would you apply the algorithm to an RGB image? Discuss your answer.